

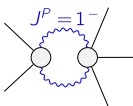
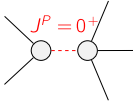
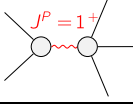
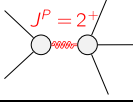
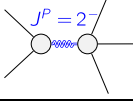
$\mathcal{K}_{0^+}^{\#1} \dagger$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1} \alpha\beta\chi$	$\mathcal{K}_{1^+}^{\#1} \alpha\beta$	$\mathcal{K}_{1^+}^{\#2} \alpha\beta$	$\mathcal{K}_{1^-}^{\#1} \alpha$	$\mathcal{K}_{1^-}^{\#2} \alpha$
	$\frac{1}{4} (6 \, Y_1 - Y_2 \, k^2)$	0				
	0	$Y_3 \, k^2$				
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{4} (Y_4 \, k^2 + 2 \, {}^{(2)}K_1)$	$\frac{-Y_4 \, k^2 - 2 \, {}^{(2)}K_1}{2 \, \sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{-Y_4 \, k^2 - 2 \, {}^{(2)}K_1}{2 \, \sqrt{2}}$	$\frac{1}{2} (Y_4 \, k^2 + 2 \, {}^{(2)}K_1)$	0	0		
$\mathcal{K}_{1^-}^{\#1} \dagger^\alpha$	0	0	$\frac{1}{2} (Y_5 + Y_6 \, k^2)$	$\frac{Y_7 \, k^2 + 2 \, {}^{(2)}K_3}{2 \, \sqrt{2}}$		
$\mathcal{K}_{1^-}^{\#2} \dagger^\alpha$	0	0	$\frac{Y_7 \, k^2 + 2 \, {}^{(2)}K_3}{2 \, \sqrt{2}}$	$\frac{1}{2} (Y_8 + Y_9 \, k^2)$		
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta$	$\mathcal{K}_{2^-}^{\#1} \alpha\beta\chi$	
				$\frac{1}{4} (Y_{10} \, k^2 + 6 \, {}^{(2)}K_1)$	0	
			$\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{2} (Y_{11} \, k^2 + 3 \, {}^{(2)}K_1)$	

$\mathcal{J}_{0^+}^{\#1} \dagger$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1} \alpha\beta\chi$	$\mathcal{J}_{1^+}^{\#1} \alpha\beta$	$\mathcal{J}_{1^+}^{\#2} \alpha\beta$	$\mathcal{J}_{1^-}^{\#1} \alpha$	$\mathcal{J}_{1^-}^{\#2} \alpha$
	$\frac{1}{\text{Det}(0^+)}$	0				
	0	$\frac{1}{k^2 ({}^{(4)}K_{15} - {}^{(4)}K_{16})}$				
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{3 \, \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \, \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\sqrt{2}}{3 \, \text{Det}(1^+)}$	$\frac{2}{3 \, \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^-}^{\#1} \dagger^\alpha$	0	0	$\frac{Y_8 + Y_9 \, k^2}{2 \, \text{Det}(1^-)}$	$\frac{-Y_7 \, k^2 - 2 \, {}^{(2)}K_3}{2 \, \sqrt{2} \, \text{Det}(1^-)}$		
$\mathcal{J}_{1^-}^{\#2} \dagger^\alpha$	0	0	$\frac{-Y_7 \, k^2 - 2 \, {}^{(2)}K_3}{2 \, \sqrt{2} \, \text{Det}(1^-)}$	$\frac{Y_5 + Y_6 \, k^2}{2 \, \text{Det}(1^-)}$		
			$\mathcal{J}_{2^+}^{\#1} \alpha\beta$	$\mathcal{J}_{2^-}^{\#1} \alpha\beta\chi$		
	$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0			
	$\mathcal{J}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^-)}$			

Abbreviations used in matrices

$Y_1 == {}^{(2)}K_1 - {}^{(2)}K_3$ && $Y_2 == 6 \, {}^{(4)}K_1 - 8 \, {}^{(4)}K_{15} + 2 \, {}^{(4)}K_{16} + 6 \, {}^{(4)}K_2 + 4 \, {}^{(4)}K_3 - {}^{(4)}K_4$ && $Y_3 == {}^{(4)}K_{15} - {}^{(4)}K_{16}$ &&
$Y_4 == 2 \, {}^{(4)}K_{16} - {}^{(4)}K_4$ && $Y_5 == 3 \, {}^{(2)}K_1 - 2 \, {}^{(2)}K_3$ && $Y_6 == 2 \, {}^{(4)}K_{15} + {}^{(4)}K_{16} - 2 \, {}^{(4)}K_2$ && $Y_7 == 2 \, {}^{(4)}K_2 + {}^{(4)}K_7$ &&
$Y_8 == 3 \, {}^{(2)}K_1 - {}^{(2)}K_3$ && $Y_9 == 3 \, {}^{(4)}K_{16} - {}^{(4)}K_2 - {}^{(4)}K_3 - 2 \, {}^{(4)}K_4 - {}^{(4)}K_7$ && $Y_{10} == 8 \, {}^{(4)}K_{15} - 2 \, {}^{(4)}K_{16} - 4 \, {}^{(4)}K_3 + {}^{(4)}K_4$ &&
$Y_{11} == 2 \, {}^{(4)}K_{15} + {}^{(4)}K_{16}$ && $\text{Det}(0^+) == \frac{1}{4} (k^2 (-6 \, {}^{(4)}K_1 + 8 \, {}^{(4)}K_{15} - 2 \, ({}^{(4)}K_{16} + 3 \, {}^{(4)}K_2 + 2 \, {}^{(4)}K_3) + {}^{(4)}K_4) + 6 \, ({}^{(2)}K_1 - {}^{(2)}K_3))$ &&
$\text{Det}(1^+) == \frac{3}{4} (k^2 (2 \, {}^{(4)}K_{16} - {}^{(4)}K_4) + 2 \, {}^{(2)}K_1)$ && $\text{Det}(2^+) == \frac{1}{4} (k^2 (8 \, {}^{(4)}K_{15} - 2 \, {}^{(4)}K_{16} - 4 \, {}^{(4)}K_3 + {}^{(4)}K_4) + 6 \, {}^{(2)}K_1)$

Lagrangian

$\begin{aligned} & \overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} + \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(2)}{\kappa}_3 \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\delta}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha} + \overset{(4)}{\kappa}_2 \partial_\chi \mathcal{K}^{\delta}_{\beta\delta} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha} + \\ & \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\alpha\chi} + \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + 2 \overset{(4)}{\kappa}_{16} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + \\ & \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} - \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + \overset{(4)}{\kappa}_7 \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha} \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} - \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \\ & \overset{(4)}{\kappa}_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$2 \mathcal{J}^{\#1\alpha\beta}_{1^+} + \mathcal{J}^{\#2\alpha\beta}_{1^+} == 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$	
Total # constraint(s):	3		
Unresolved pole(s)	# d.o.f.	Pole structure(s)	
	6	$\begin{aligned} & k^4 (6 \overset{(4)}{\kappa}_{16}^2 + 4 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 + 8 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_4 + \\ & 4 \overset{(4)}{\kappa}_{15} (3 \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - 2 \overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_7) - \overset{(4)}{\kappa}_7^2 - 2 \overset{(4)}{\kappa}_{16} (7 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + 2 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_7)) + 18 \overset{(2)}{\kappa}_1 (\overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_3) + \\ & 2 k^2 (-3 (-4 \overset{(4)}{\kappa}_{16} + 3 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + 2 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_7) \overset{(2)}{\kappa}_1 + \overset{(4)}{\kappa}_{15} (6 \overset{(2)}{\kappa}_1 - 2 \overset{(2)}{\kappa}_3) + (-7 \overset{(4)}{\kappa}_{16} + 2 \overset{(4)}{\kappa}_3 + 4 \overset{(4)}{\kappa}_4) \overset{(2)}{\kappa}_3) \end{aligned}$	
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{6 (\overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_3)}{6 \overset{(4)}{\kappa}_1 - 8 \overset{(4)}{\kappa}_{15} + 2 \overset{(4)}{\kappa}_{16} + 6 \overset{(4)}{\kappa}_2 + 4 \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4}$	$-\frac{4}{6 \overset{(4)}{\kappa}_1 - 8 \overset{(4)}{\kappa}_{15} + 2 \overset{(4)}{\kappa}_{16} + 6 \overset{(4)}{\kappa}_2 + 4 \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4}$
	3	$\frac{2 \overset{(2)}{\kappa}_1}{-2 \overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_4}$	$\frac{4}{6 \overset{(4)}{\kappa}_{16} - 3 \overset{(4)}{\kappa}_4}$
	5	$-\frac{6 \overset{(2)}{\kappa}_1}{8 \overset{(4)}{\kappa}_{15} - 2 \overset{(4)}{\kappa}_{16} - 4 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4}$	$\frac{4}{8 \overset{(4)}{\kappa}_{15} - 2 \overset{(4)}{\kappa}_{16} - 4 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4}$
	5	$-\frac{3 \overset{(2)}{\kappa}_1}{2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$	$-\frac{2}{2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		